

Application of STEM principles and practices

Communication

This project was taken on by a larger team of 13 individuals split into 3 subgroups. Our subgroup's work involved making a carrying box compatible with the routine medical applications on Komodo dragons by zookeepers (ultrasound of females and X-ray imaging). The other two groups' work centered around frame design and the connection of multiple boxes.

Because of this collaborative aspect of the project, communication across teams is a critical STEM principle we instituted. Design choices of each team influences the overall project's adherence to every other team's criteria and design requirements. Thus, communication permitted the overall product generated by the entire team to match the design requirements. Likewise, funding was given to each team individually. As a consequence of this, our larger team instituted the 2 following collaborative measures to streamline product development.

(1) **Weekly meetings.** Every week, each team's changing objectives, criteria, and designs were communicated to each team. This ensured that as each team's designs iterated, other teams' objectives and criteria were not impeded.

Meetings likewise allowed for burden sharing and accountability. The needs of one team were often contingent on the work of other teams. Weekly meetings allow for each group to ensure other groups get work done, helping the team adhere to the timeline.

During these meetings, design requirements outlined by each team were discussed, communicated, and incorporated into team requirements to ensure consistency in goals. Since certain groups' designs are more critical to adhering to design requirements than others, meetings allow for distributed responsibility in adhering to a common set of design requirements. For instance, the frame-design team focuses more on enhancing the weight-load capacity of the box than another group may, since the frame design has the largest impact on that design requirement. At the same time, each team communicates with the frame-design team to ensure this criterion is met, since they have a vested interest in meeting that design requirement.

In this way, these meetings serve as a prerequisite to meeting design requirements by ensuring each group adheres to requirements and by distributing responsibility for adhering to these requirements.

(2) **Fund sharing.** Because the objectives, problems, and solutions for each team differed, the amount that each team would spend likewise differ. Thus, the even distribution of funds to each team would lead to certain teams being underfunded while others were overfunded in their objectives.

Consequently, the team decided to share funds, with each team communicating in weekly meetings (and whenever appropriate) regarding how much funding their design would require. In these meetings, decisions to allocate funds for a group's material were discussed and agreed upon before purchasing materials.

Compliant Mechanisms

Compliant mechanisms were used in our design to create spring latches that securely hold the side door open or closed. As opposed to traditional mechanisms that rely on multiple rigid components (e.g. springs or pins), the flexible structure of compliant mechanisms in and of themselves are used to achieve motion and force.

The use of compliant mechanisms provided 3 distinct benefits:

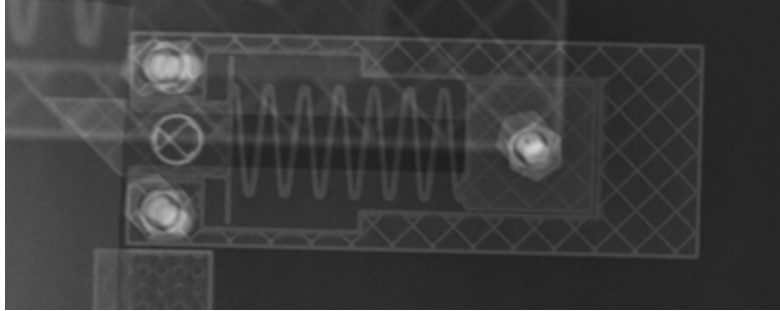
(1) **Reliability.** Typical latching mechanisms to hold open the door would be constructed out of metal and would involve multiple components, as shown below.



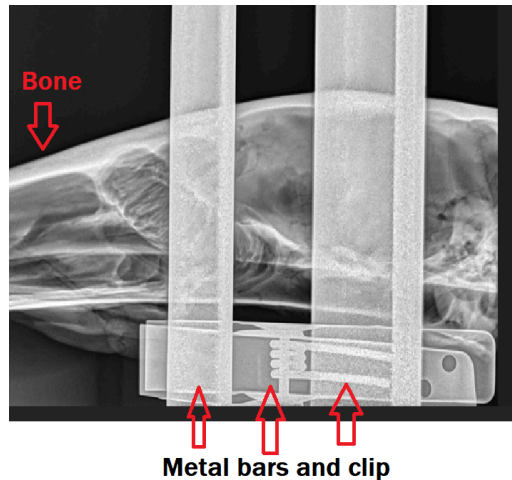
The complex, multi-part design of typical latches and hinges creates greater points of failure for the mechanism, resulting in poorer long-term reliability (a requirement for the design). Since compliant mechanisms rely on their very structure to achieve their attained goal (i.e. holding a door in place), there are fewer points of failure presented with compliant mechanisms in contrast.

On top of the increase in reliability, the fewer parts needed with compliant mechanisms improves the solution's simplicity of latching. Only a spring must be pulled back to unlatch a door, whereas a typical metal latch requires a multi-step process of rotating the latch, translating it, and rotating it again.

(2) **X-ray compatibility.** The metal used in existing solutions (i.e. hinges and latches) would deflect X-rays and appear as artifacts on X-ray images taken of the dragon. Since the latches must latch onto the door, which itself is between the dragon and the X-ray, these artifacts reduce the insights provided by X-ray imaging. The simple design and PLA plastic in the compliant mechanism reduce artifacting, as shown below.



X-ray artifacting from compliant latching mechanism in front of empty space(above).



X-ray artifacting from metal in front of a bone (above).

(3) **Production cost.** Metal hinges or latches cost relatively more than compliant mechanisms because they are typically made of more expensive materials and are more complex. Our team is able to 3D print compliant mechanisms at a low cost, providing a relatively economical alternative to hinges.

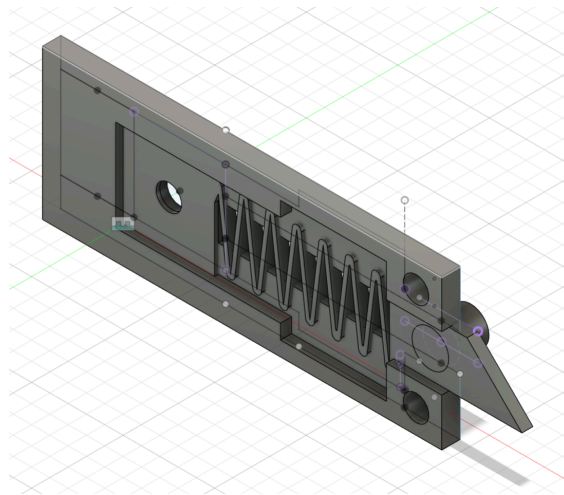
Biomedical engineers and zookeepers were consulted to ensure that compliant mechanisms would both (a) not interfere with results of medical imaging and (b) provide a simple, reliable solution.

CAD

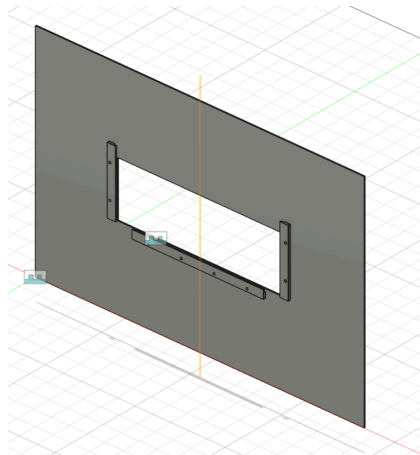
To reduce reliance on resource intensive physical prototypes and testing, our group opted to use computer-aided design (CAD) to design prototypes. Furthermore, CAD allows the team to model, export, and 3D print prototypes to test. In this, CAD permits the team to create more precise prototypes that emulate the quality of the final product at a low cost. CAD therefore provides a quality way to cut costs and streamline the team's workflow.

CAD further assists in the engineering principle of iteration. By enabling the team to 3D print designs quickly and cheaply, CAD promotes iteration by streamlining the creation of prototypes and consequent revision of prototypes via adjustments on the software.

More specifically, CAD allows the team to model complex geometries for designs that keep the ultrasound door closed. In doing so, CAD permits the team to ensure that the latches holding the ultrasound door closed were feasible and effective. Our group was able to make modifications to the casing of these latching mechanisms when the latches were unable to support the weight of the door through CAD, demonstrating that CAD assisted in the iterative approach to the group's solutions.



Through modeling the track system and the overall design of the side panel, our team is likewise able to plan out the precise location of parts relative to each other. This reduces the time it takes to assemble the prototype once materials arrive, permitting quick iteration and the production of a well-planned prototype.



The prototype made through CAD and design iterations were reviewed by engineers. Their input led to modifications that would allow the design to better fit the criteria, such as by making the design more X-ray compatible.

Interdisciplinary Work

Throughout the project, our team has leveraged expertise from multiple disciplines to ensure that the crate met all technical and medical requirements outlined.

Design requirements necessitating the side panel and ultrasound-door assembly be X-ray compatible shape our design from a biomedical perspective. Likewise, the requirements calling for a reliable ultrasound door as well as durable supports for the door necessitate the group approach the problem and solution through a mechanical engineering perspective.

Because of this multifaceted, multidisciplinary nature of our problem and solution, the expertise the group draws from is likewise diverse.

Biomedical engineers specializing in medical imaging are consulted to advise what materials would allow the design to be X-ray compatible, shaping our initial approach. Any modifications in our design were checked to ensure they remained X-ray compatible.

Mechanical and biomedical engineers likewise are consulted to ensure that the ultrasound door's latching mechanism is reliable and durable. Mechanical engineers have likewise provided suggestions for making the ultrasound door track more durable and resilient to force.

The team presents prototypes of the solution to zoo veterinarians--the primary stakeholders--to understand how the solution will be used. This insight is used (a) to shape iterations of the design to suit the needs of end users and (b) refine the group's problem statement by communicating what issues veterinarians run into.

In this way, the multidisciplinary approach ensures blindspots in our design do not form. Were the group to draw from a limited base of expertise, the implications of making or not making certain design choices could be overlooked, making it more difficult for the team to adhere to the design requirements and criteria. As such, the interdisciplinary nature of our approach permits the team to achieve the best solution possible.